

Panel MS-AR(1) from 1995: common RW trend and RE intercepts

Seasonally adjusted real GDP per worker

1 Model

Sample starts in 1995Q1. Two-step. Step 1 is a common trend from average growth of countries observed in consecutive quarters:

$$\Delta\tau_t = \frac{1}{n_t^{\text{cont}}} \sum_{i \in C_t \cap C_{t-1}} (y_{it} - y_{i,t-1}), \quad (1)$$

$$\tau_t = \tau_{t_0} + \sum_{s=t_0+1}^t \Delta\tau_s. \quad (2)$$

The level of τ is pinned to the balanced-core mean at t_0 . Mean $\Delta\tau$ is 0.0041 per quarter (3 countries present in every quarter). Step 2 is estimated on $y_{it}^* = y_{it} - \hat{\tau}_t$:

$$y_{it}^* = a_i + z_{it}, \quad (3)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2), \quad (4)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it})z_{it} + \sigma(s_{it})\varepsilon_{it}. \quad (5)$$

No catch-up term. No extra gt . Three regimes; median μ pinned at 0; σ and ρ switch; $|\rho| < 0.99$. Random intercepts: 11-point Gauss–Hermite. Plotted cycles use posterior-mean a_i . τ_t is discarded for the structural MS-AR. Standard errors ignore step-1 uncertainty.

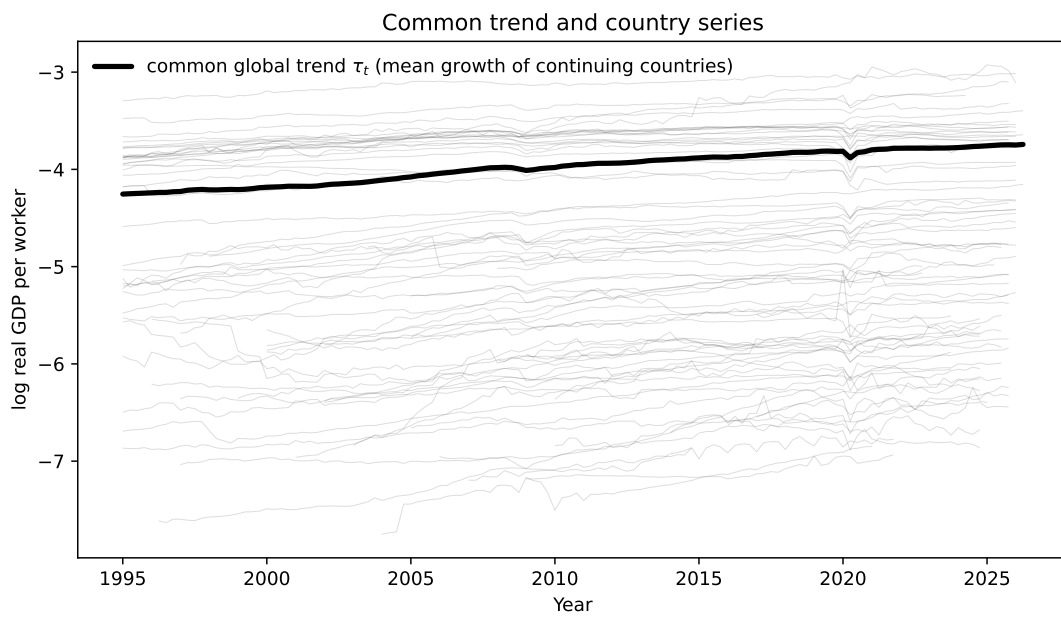


Figure 1: Faint lines: country log real GDP per worker. Thick line: common τ_t from average growth of continuing countries.

2 RE intercepts around the common RW trend (no catch-up)

2026-09-17 SA panel, $y_{it} = a_i + \tau_t + z_{it}$, from 1995. 74 countries, 7648 observations, 1995-Q1–2026-Q2. $|\rho| < 0.99$. Fitted countries: 74. Observations: 7648. Log-likelihood: 20676.70. Ergodic mean of the cycle $E[z] = 0.0694$ (0.0236). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -0.6457$ (0.0114), $\omega = 0.6738$ (0.0072). Posterior-mean a_i : mean -1.155, min -3.298, max 0.618.

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.5317 (0.1067)	0.0000 —	0.2928 (0.0260)
σ	0.0719 (0.0029)	0.0074 (0.0001)	0.0190 (0.0005)
ρ	0.9897 (0.0039)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.6740 (0.0375)	0.0000 (0.0001)	0.3260 (0.0375)
Π_1	0.0095 (0.0032)	0.9438 (0.0056)	0.0467 (0.0062)
Π_2	0.0622 (0.0082)	0.0676 (0.0077)	0.8701 (0.0112)
π	0.0930 (0.0103)	0.4954 (0.0258)	0.4116 (0.0221)

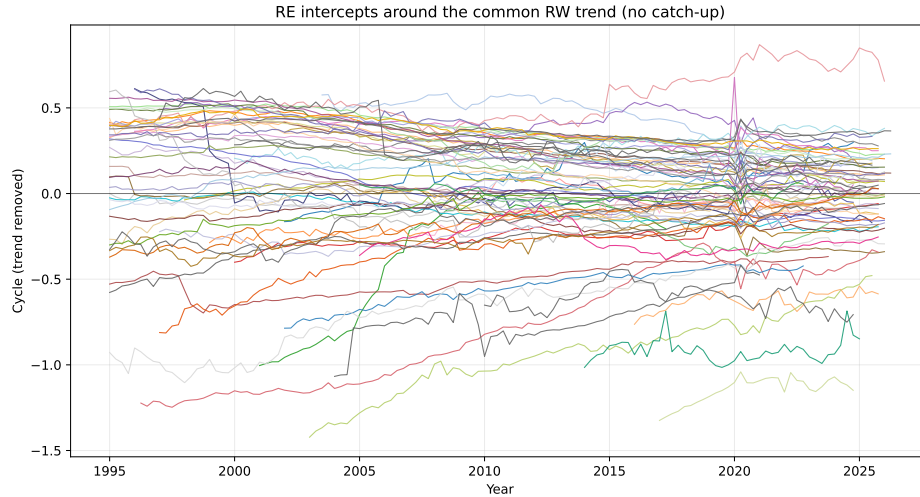


Figure 2: Country cycles after removing τ_t and posterior-mean a_i .